ContextKeeper WP-004: Open infrastructure and AI toolkit reference (March 2026)

This technical reference captures the state of open-weight models, inference infrastructure, agent frameworks, and supporting tooling as of March 18, 2026. The open AI ecosystem has undergone dramatic shifts since the original whitepaper: Meta launched its first MoE-architecture models (Llama 4), OpenAI released its first open weights since GPT-2, the Model Context Protocol became a Linux Foundation standard, and GPU cloud pricing collapsed 64–75% from peak. Introl Every section below provides verified specifications, benchmarks, licensing, and deployment requirements—with explicit uncertainty flags where claims cannot be independently confirmed.

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1. Open-weight model landscape

1.1 Meta Llama 4 family: MoE architecture arrives

Meta released **Llama 4 Scout** and **Llama 4 Maverick** on **April 5, 2025**— Meta its first Mixture-of-Experts models, replacing the dense architecture used in all prior Llama generations. Both use a novel **iRoPE** attention architecture with interleaved dense and MoE layers, APXML and both are natively multimodal via early fusion Meta with a MetaCLIP-based vision encoder.

Llama 4 Scout (17B active / 109B total, 16 experts) DEV Community Wikipedia delivers a **10-million-token context window**— Meta APXML the longest of any open-weight model. Meta Despite activating only 17B parameters per forward pass, Scout matches or exceeds Llama 3.1 70B on most benchmarks and surpasses Llama 3.1 405B on MMLU-Pro (instruct: **74.3** vs 73.4) and GPQA Diamond (instruct: **57.2** vs 49.0). It fits on a single H100 with INT4 quantization Meta APXML (~67 GB in Q4_K_M). Ollama On consumer hardware, Unsloth's 1.78-bit GGUF quantization fits in ~24 GB VRAM at ~20 tok/s— Unsloth Unsloth AI tight but usable on an RTX 4090.

Llama 4 Maverick (17B active / ~400B total, 128 experts) Meta Wikipedia pushes further with a **1-million-token context window**. OpenRouter Maverick significantly outperforms Llama 3.1 405B across all reported benchmarks—MMLU **85.5** (vs 85.2), MATH **61.2** (vs 53.5), GPQA Diamond **69.8** (vs 49.0)—while activating only 17B parameters versus 405B dense. Meta claims it beats GPT-4o and Gemini 2.0 Flash broadly. Meta However, Q4_K_M quantization requires ~245 GB, making it a datacenter model InsiderLlm (fits on 8×H100 in FP8). On consumer hardware, only extreme 1.78-bit quantization (~122 GB) is feasible across 2×48 GB GPUs. Unsloth Unsloth AI

⚠️ **Benchmark controversy**: Meta submitted an undisclosed "experimental" Maverick variant optimized for conversationality to LMArena, achieving ELO 1417. This was not the publicly

released model. LMArena criticized Meta's transparency, and the community flagged this as benchmark manipulation. Independent evaluations suggest Gemini 2.5 Pro and Claude 3.7 Sonnet significantly outperform both Scout and Maverick on reasoning-intensive tasks like GPQA Diamond and LiveCodeBench.

Llama 4 Behemoth (288B active / ~2T total, 16 experts) was announced (Wikipedia) alongside Scout and Maverick as a teacher model (Meta) used for co-distillation. (DataCamp) It remains **unreleased as of March 2026**. (Skywork) Originally expected at LlamaCon (April 29, 2025), it was postponed to June 2025, then fall 2025, and then indefinitely. WSJ reported internal concerns about performance not matching claims. (SiliconANGLE) No formal cancellation has been announced, (Aicerts News) but no release timeline exists. (Aicerts News)

Both Scout and Maverick are available in **Ollama** (`ollama run llama4:scout`, `ollama run llama4:maverick`) with Q4_K_M, Q8_0, and FP16 quantizations. All Llama 4 models use the **Llama 4 Community License Agreement** (Hugging Face) (Hugging Face) (commercial use permitted, 700M MAU threshold, (Hugging Face) "Built with Llama" attribution required, not OSI open-source). (huggingface) (Skywork) Training consumed **7.38M GPU hours** on H100-80GB (Hugging Face) (Scout: 5.0M, Maverick: 2.38M). (huggingface)

1.2 Llama 3.3 and 3.2 remain essential

Llama 3.3 70B (released **December 6, 2024**) is a dense 70B-parameter instruction-tuned model (Hugging Face) with a **128K context window** (huggingface) that delivers near-parity with Llama 3.1 405B at one-sixth the parameter count. It actually **surpasses** the 405B model on instruction following (IFEval: **92.1** vs 88.6), reasoning (GPQA Diamond: **50.5** vs 49.0), and math (MATH: **77.0** vs 73.8). (huggingface) It is **the recommended 70B-class model** for local deployment, competitive with Qwen 2.5 72B. In Q4_K_M quantization (~42.5 GB), it fits on a single 48 GB GPU (A6000, L40S); (Hugging Face) on an RTX 4090 with CPU offloading at moderate context lengths. (Llama 3.2) (Novita AI) Available on Ollama as `ollama run llama3.3`. This was confirmed as the final Llama 3-series release. (Hacker News)

Llama 3.2 (released **September 25, 2024**) spans four sizes: **1B and 3B** (text-only, (Meta) created via pruning + distillation from Llama 3.1 8B) and **11B and 90B** (multimodal vision models, built via late-fusion cross-attention on top of Llama 3.1 8B and 70B respectively). (meta) All support **128K context**. (InfoQ) The 90B Vision model matches GPT-4o on ChartQA and beats Claude 3 Opus on AI2D. (Medium) (IBM) EU licensing restrictions apply to the multimodal variants. (Hugging Face) All sizes are available on Ollama. (Ollama)

1.3 Mistral Large 3 becomes the open-weight flagship

Mistral AI's current flagship is **Mistral Large 3**, released **December 2, 2025**—a sparse MoE model with **675B total / 41B active parameters**, **256K context window**, (IntuitionLabs) multimodal (text + images), and released under **Apache 2.0**. (eWEEK) This is Mistral's strongest open-weight

offering and its first MoE since the original Mixtral. Microsoft Azure describes it as "the strongest fully open model developed outside of China." Microsoft Azure

The December 2025 release also included **Ministral 3** (dense models at 3B, 8B, and 14B, Wikipedia all Apache 2.0) and **Devstral 2** (coding-specialized, 24B). Earlier 2025 releases included Mistral Small 3.1 (March), Mistral Medium 3 (May), and the Magistral reasoning models (June). Wikipedia Mistral remains firmly committed to open-weight releases—it has not shifted to API-only.

1.4 Qwen 3 and 3.5 lead the Chinese open-weight ecosystem

Qwen 3 launched **April 28, 2025** with eight model sizes from 0.6B to 235B, including two MoE variants. Alibaba Cloud Community TechCrunch The flagship **Qwen3-235B-A22B** (235B total / 22B active, Ollama 128 experts, 8 active per token) arXiv competes head-to-head with DeepSeek-R1, o1, and Gemini 2.5 Pro. Ollama Qwen All Qwen 3 models are **Apache 2.0**. Qwen Training used **36 trillion tokens** across 119 languages. Alibaba Cloud Community Qwen A key efficiency result: Qwen3-4B rivals Qwen 2.5-72B-Instruct performance. Ollama Qwen

Alibaba rapidly iterated beyond Qwen 3: **Qwen3-Next** (September 2025, 80B/3B active with hybrid Gated Delta Networks + sparse MoE), NVIDIA Developer Alibaba Cloud Community **Qwen3-Coder** (480B/35B active), Ucstrategies News **Qwen3-Omni** (multimodal: text, images, audio, video), Wikipedia and **Qwen 3.5** (February 17, 2026, 397B parameters, native multimodal, 201 languages). CNBC The **Qwen3.5-35B-A3B** model reportedly surpasses Qwen3-235B using only 3B active parameters via 256 experts with Gated Delta Networks. VentureBeat ⚠️ Many benchmark claims are self-reported by Alibaba and not fully independently verified. CNBC

1.5 Google Gemma 3 and Microsoft Phi-4

Gemma 3 (released **March 12, 2025**) offers 1B, 4B, 12B, and 27B sizes with **128K context**. Hugging Face The 4B+ sizes are multimodal (text + image via SigLIP encoder). Hugging Face Gemma 3 27B achieved **Chatbot Arena ELO 1338** at launch—top-10 globally, surpassing many proprietary models. Libertify License is the **Gemma Terms of Use** (permissive commercial use but more restrictive than Apache 2.0). Available on Ollama. Gemma 2 (2B, 9B, 27B, 8K context) is effectively superseded. Ollama

Phi-4 (released **December 12, 2024**) is a 14B dense model Ollama with **16K context** under **MIT License**, Artificial Analysis specializing in math and reasoning Microsoft Community Hub (MATH: **80.4%**, outperforming GPT-4o). Awesome Agents The Phi-4 family expanded rapidly: **Phi-4-mini** (3.8B, SiliconANGLE 128K context, Feb 2025), Ollama **Phi-4-multimodal** (5.6B, vision + audio, Feb 2025), SiliconANGLE Microsoft Community Hub **Phi-4-reasoning** (14B, CoT reasoning trained on o3-mini demonstrations, Microsoft May 2025), Hugging Face and most recently **Phi-4-reasoning-vision-15B** (March 4, 2026, multimodal reasoning). Hugging Face All MIT License. Microsoft Azure Phi-4-reasoning outperforms o1-mini and is competitive with full DeepSeek R1 on AIME 2025—remarkable for a 14B model. Microsoft

1.6 DeepSeek-R1 matches frontier reasoning

DeepSeek-R1 (released **January 20, 2025**) Fireworks AI is a 671B MoE model ollama (37B active, 256 routed experts per block, 8 active per token) Epoch AI with **128K context** under **MIT License**. It matches OpenAI o1 on reasoning: AIME 2024 **79.8%** DEV Community (vs o1's 79.2%), MATH-500 **97.3%** Fireworks AI (vs 96.4%). Medium It significantly outperforms GPT-4o and Claude 3.5 Sonnet on mathematical reasoning. The **R1-0528 update** (May 2025) improved Ollama Ollama AIME 2025 from 70% to 87.5%. Hugging Face Six distilled dense models (1.5B to 70B) provide accessible alternatives— Built In Ollamathe 32B distill matches o1-mini performance. Medium

The full 671B model requires substantial hardware: ~336– DEV Community350 GB in Q4 quantization (4×H100), Aidevdayindia Geeky Gadgets though 1.58-bit extreme quantization fits in ~140 GB. Ollama Available on Ollama with sizes from 1.5B distills to the full 671B (404 GB download in Q4_K_M).

⚠️ **Significant controversies** surround DeepSeek: data transmitted to China Mobile servers, Feroot Security government bans in Italy, US Navy, NASA, Feroot Security South Korea, CNBC Australia, and Taiwan. TechTarget In February 2026, Anthropic accused DeepSeek of using fraudulent accounts to train on millions of Claude conversations. Wikipedia Self-hosting open weights eliminates cross-border data flow but model-level censorship patterns persist. IAPP

1.7 OpenAI's first open weights: GPT-OSS

GPT-OSS-120B and **GPT-OSS-20B** Hyperstack were released **August 5, 2025**—OpenAI's first open-weight models since GPT-2 Hyperstack Medium in 2019. Both are MoE architectures under **Apache 2.0**. OpenAI

GPT-OSS-120B (116.8B total / **5.1B active** per token, Hugging Face InfoQ 128K context) matches or exceeds o4-mini on competition coding (Codeforces), MMLU, and tool calling. OpenAI It runs on a **single 80 GB GPU** OpenAI (H100 or MI300X) Hugging Face with native MXFP4 quantization (~61 GB). AMD GPT-OSS-20B (20.9B total / **3.6B active**, InfoQ 128K context) matches o3-mini on most benchmarks and runs on just **16 GB of memory**— OpenAIconsumer hardware viable. Both are available on Ollama (`ollama pull gpt-oss:120b`, `ollama pull gpt-oss:20b`), Ollama Hugging Face, vLLM, and llama.cpp. Safety-tuned variants (gpt-oss-safeguard) were released October 2025. OpenAI

1.8 The MoE landscape beyond Mixtral

The open-weight MoE ecosystem has expanded significantly:

- **DBRX** (Databricks, March 2024): 132B/36B active, fine-grained 16-expert routing, Databricks Substack Databricks Open Model License
- **Grok-1** (xAI, March 2024): 314B/~76B active, Nestify Apache 2.0, but only raw base model (no instruction tuning) Grokipedia

- **Jamba 1.5** (AI21, August 2024): Hybrid SSM (Mamba) + Transformer + MoE, AI21 Labs 398B/94B active (Large), 256K context— Analytics Vidhya unique architecture combining structured state-space models with attention NVIDIA Developer
- **Snowflake Arctic** (April 2024): 480B/17B active, dense-MoE hybrid, GitHub Apache 2.0, enterprise-task focused VentureBeat
- **OLMoE** (AI2, 2024): 7B/1B active, fully open (data, code, weights, training logs), Apache 2.0
- **Phi-3.5-MoE** (Microsoft): ~42B/~6.6B active, MIT License

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2. Model Context Protocol as open infrastructure

2.1 MCP architecture and specification

The **Model Context Protocol (MCP)** is an open protocol created by **Anthropic** (announced November 2024) that standardizes how LLM applications connect to external data sources and tools. Inspired by the Language Server Protocol (LSP), MCP defines a JSON-RPC 2.0-based client/server architecture. Model Context Protocol In **December 2025**, Anthropic donated MCP to the **Agentic AI Foundation** under the **Linux Foundation**, co-founded by Anthropic, Block, and OpenAI. Wikipedia The latest specification version is **2025-11-25**.

MCP uses a three-tier architecture. **MCP Hosts** (Claude Desktop, Cursor, VS Code Copilot) are AI applications that orchestrate access to context. **MCP Clients** maintain 1:1 connections to servers, handling JSON-RPC formatting, session management, and capability negotiation. IBM **MCP Servers** expose three core primitives: **Tools** (functions the model can execute), **Resources** (data the application can access), and **Prompts** (templated workflows for users). Model Context Protocol Communication always flows Host → Client → Server.

Two transport layers are supported. **stdio** launches the server as a subprocess with JSON-RPC over stdin/stdout—ideal for local integrations Hashnode with microsecond-level latency. MCPcat **Streamable HTTP** (introduced March 2025, replacing legacy HTTP+SSE) provides a single HTTP endpoint supporting both POST (JSON or SSE responses) and GET, with session management via `Mcp-Session-Id` headers and resumable streams. Fka For remote servers, authentication uses **OAuth 2.1 with PKCE**, with mandatory Resource Indicators (RFC 8707) binding tokens to specific servers and Protected Resource Metadata Auth0 (RFC 9728) for auth discovery.

Tools are defined using JSON Schema with `name`, `description`, `inputSchema`, Modelcontextprotocol and optional `outputSchema` (added June 2025). The November 2025 spec added `ToolAnnotations` with behavioral hints WorkOS (`readOnlyHint`, `destructiveHint`, `idempotentHint`) Model Context Protocol and introduced Tasks for async operations, WorkOS enhanced sampling with tool calling, and the MCP Extensions system. Modelcontextprotocol

2.2 SDKs span ten languages across three tiers

Official SDKs follow a tiered maturity system: Model Context Protocol

- **Tier 1** (full spec compliance, fast maintenance): **TypeScript** (`@modelcontextprotocol/sdk` on npm), GitHub **Python** (`mcp` on pip, includes FastMCP high-level framework), **C#/.NET** (maintained with Microsoft), GitHub **Go**
- **Tier 2** (good compliance, active maintenance): **Java** (Spring AI integration), Spring **Rust**
- **Tier 3** (basic functionality): **Swift**, **Ruby**, **PHP** (maintained with The PHP Foundation), **Kotlin** (1,297 stars) GitHub

Over **17,967 repositories** are tagged with MCP on GitHub (Python: 6,013, TypeScript: 5,132). GitHub The SDK ecosystem has surpassed **97 million cumulative downloads** in its first year. Shareuhack Building a basic MCP server requires roughly 20 lines of Python using FastMCP or comparable TypeScript with the official SDK.

2.3 Registry and ecosystem scale

The **official MCP Registry** (registry.modelcontextprotocol.io, launched September 2025) Modelcontextprotocol operates as a metaregistry storing server metadata in `server.json` format, referencing npm, PyPI, Docker Hub, and GitHub Releases for actual packages. It uses reverse-DNS namespacing and GitHub OAuth for publisher authentication. As of March 2026, the official registry contains ~**87 validated servers**, Descope while the broader ecosystem includes **10,000+** meaningful implementations (per PulseMCP) Shareuhack and **14,274** total listings (per MCP.so). Descope Notable servers include GitHub (20,200+ stars, 100+ tools), Playwright (browser automation), Slack, PostgreSQL, Google Cloud (AlloyDB, BigQuery, Spanner), Figma, Shareuhack Stripe, and Salesforce.

2.4 MCP in local inference engines and security posture

No major local inference engine natively implements MCP. Ollama, vLLM, TGI, and LocalAI all serve as LLM backends within MCP host applications—they provide function/tool calling capability and OpenAI-compatible APIs, which MCP hosts use to generate structured tool calls. Community bridges like `ollama-mcp-bridge`, GitHub `MCPHost` (Go), Hashnode ClickHouse and `mcp-client-for-ollama` GitHub connect Ollama to MCP servers. GitHub Medium The architecture is intentionally layered: inference engines handle generation, while MCP handles protocol-level tool orchestration.

MCP security risks are substantial and well-documented. **Invariant Labs** (April–May 2025) demonstrated Tool Poisoning Attacks that exfiltrated SSH keys from Cursor, cross-repository data leakage via GitHub MCP (affecting the 20,200-star official server), and WhatsApp history exfiltration via malicious MCP servers. Supply chain attacks were demonstrated against the Smithery platform and `mcp-remote` (CVE-2025-6514). Shareuhack Palo Alto Unit 42 documented

sampling-based attacks enabling resource theft and conversation hijacking. Best practices include **principle of least privilege** on tokens, tool allowlisting per workflow, MCP Gateway patterns for centralized enforcement, egress controls, tool description monitoring for "rug pull" attacks, and comprehensive audit logging.

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3. Agent framework convergence

3.1 Microsoft Agent Framework succeeds AutoGen and Semantic Kernel

Microsoft announced the **Microsoft Agent Framework** in public preview on **October 1, 2025**, Microsoft describing it as "the next generation of both Semantic Kernel and AutoGen" created by the same teams. Microsoft Learn It reached **RC2** status in early 2026 with GA targeted for **Q1 2026** (end of March). Cloudsummit Both AutoGen and Semantic Kernel are now in maintenance mode (bug fixes and security patches only). Microsoft Agent Framework SDKs are available for **Python** (`pip install agent-framework --pre`) and **C#/.NET**, Microsoft Agent Framework GitHub with Java on the roadmap. The repo has **~7,200 GitHub stars**.

The framework combines AutoGen's agent orchestration (LLM-driven, creative reasoning) with Semantic Kernel's workflow orchestration (business-logic driven, deterministic). Medium Key architectural features include graph-based workflows with streaming and checkpointing, multi-provider support GitHub (Azure OpenAI, Anthropic, Ollama, AWS Bedrock), interoperability with A2A, AG-UI, and MCP protocols, Microsoft Agent Framework and enterprise features including OpenTelemetry and Entra ID authentication. Medium

3.2 Google ADK, OpenAI Agents SDK, and the broader landscape

Google Agent Development Kit (ADK) Google launched **April 9, 2025** MarkTechPost at Cloud NEXT and rapidly grew to **17,600 stars**. GitHub Available in Python Google GitHub (v1.27.2), TypeScript, Go, and Java. GitHub Key differentiators include native Vertex AI integration, A2A protocol support, PyPI multi-agent hierarchies with coordinator agents, Google built-in evaluation via `adk eval`, GitHub and bidirectional audio/video streaming. It's model-agnostic (optimized for Gemini but supports Claude, Ollama, and others via LiteLLM). Google GitHub

OpenAI Agents SDK GitHub (confirmed at **v0.12.3**, March 16, 2026) PyPI has **~20,000 stars**. GitHub Core primitives include Agents, Handoffs (task delegation between agents), Guardrails (safety checks), and Tracing (integrated with OpenAI's evaluation tools). OpenAI MCP integration is confirmed as a core feature. GitHub The SDK is provider-agnostic via the Chat Completions API interface— GitHubany compatible endpoint works, including through LiteLLM, though LiteLLM is not a built-in dependency. Mem0 It succeeds the experimental Swarm framework Mem0 and the now-deprecated Assistants API (sunset targeted mid-2026). OpenAI

Smolagents (Hugging Face) GitHub reached **v1.24.0** GitHub PyPI with **~25,500 stars**, GitHub supporting text, vision, video, and audio inputs. GitHub Recent additions include MCP support, multiple sandbox options (E2B, Docker, Modal, Pyodide+Deno), GitHub PyPI OpenTelemetry integration, and CLI tools (`smolagent`, `webagent`). Hugging Face Hugging Face

LangGraph is at **v1.1.0** (released March 10, 2026— PyPI ⚠️ not v1.1.2 as stated in the original brief) with **26,200 stars**. Version 1.1 introduced type-safe streaming and invoke (opt-in `version="v2"`), Pydantic/dataclass coercion, GitHub Deep Agents for complex planning, GitHub and AG-UI protocol support. **LangGraph Platform** provides hosted deployment with persistence and task queues. GitHub **LangGraph Studio** offers visual graph debugging (web version recommended over macOS desktop app). GitHub Note: LangGraph uses a **proprietary license**, PyPI not MIT or Apache. Used in production at LinkedIn, Uber, Medium Klarna, Replit, and 400+ companies.

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4. Inference backend updates

4.1 Ollama dominates local deployment at 162K stars

Ollama is at approximately **v0.17.x** (v0.16.1 confirmed February 12, 2026; v0.17.1 referenced in subsequent release notes) GitHub with **~162,000 GitHub stars**. The model library includes **100+ model families** spanning Llama 4, Qwen 3/3.5, DeepSeek-R1, Gemma 3, Phi-4, Mistral Large 3, GPT-OSS, and many more. Ollama **does not natively support MCP** (open feature request #7865) GitHub but supports tool/function calling since v0.3.1, enabling MCP integration through community bridges. Multimodal support includes Llama 3.2 Vision, Llama 4, Qwen3-VL, and LLaVA models. Recent features include cloud model access (`:cloud` tag without local download), ROCm 7 support, MLX engine integration for Apple Silicon, `ollama launch` for CI/CD, and partnerships with NVIDIA for DGX Spark optimization. Ollama

4.2 vLLM, llama.cpp, and SGLang form the production serving triad

vLLM (v0.17.1, **~73,500 stars**) has matured into the default production inference server. Spheron Key additions since mid-2025 include FlashAttention 4 integration, GitHub Model Runner V2 with pipeline parallelism (**30.8% throughput improvement**), GitHub disaggregated prefill/decode scheduling, a Realtime API for WebSocket streaming, GitHub RTX Blackwell (SM120) support, GitHub and Transformers v5 compatibility. Structured output (JSON schema enforcement), speculative decoding Spheron (Eagle3 with CUDA graphs), and multi-LoRA serving are all fully supported.

llama.cpp (**~94,700 stars**) continues as the foundational C/C++ inference engine. The ggml.ai team **joined Hugging Face on February 20, 2026** while keeping all projects open-source under MIT. GGUF v3 introduced importance-matrix quantization (i-quants) with non-linear codebooks superior to K-quants at extreme bit widths, Botmonster Tech plus MXFP4 support for MoE models.

Llama 4 is fully supported. NVIDIA reported a **35% throughput increase** for MoE models on llama.cpp at CES 2026. Enclaveai Key capabilities include grammar-constrained decoding (GBNF), LoRA hot-swapping, GitHub Docker Hub integration for GGUF pulls, Docker and cross-platform acceleration (CUDA, Metal, ROCm, Vulkan, SYCL, OpenCL, and extensive CPU SIMD). Wikipedia

SGLang (v0.5.9, ~**24,300 stars**) from UC Berkeley/LMSYS is now deployed on **400,000+ GPUs worldwide** by xAI, AMD, NVIDIA, LinkedIn, Cursor, and major cloud providers. GitHub Its key differentiator, **RadixAttention** (automatic KV cache reuse via radix tree), delivers up to **5–6× higher throughput** than baseline systems on structured generation, multi-call, and RAG workloads. Other differentiators include compressed finite-state machine for constrained decoding, prefill-decode disaggregation, and native RL integrations. GitHub SGLang is essential for any technical reference on inference infrastructure.

TabbyML (v0.13.0, ~33,000 stars) has **stalled development**—no major release since July 2024. The team appears to be pivoting to a new product called Pochi. GitHub TabbyML is a coding assistant (GitHub Copilot alternative), not an inference backend, Tabby and should be treated as a historical reference rather than a current recommendation.

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5. GPU cloud pricing has collapsed

5.1 Current pricing across major providers

H100 on-demand pricing has dropped **64–75% from its ~$8/hr peak** in late 2024, Introl Accio driven by 300+ new GPU cloud providers entering the market in 2025 Accio and AWS cutting H100 prices ~44% in June 2025. Introl

GPU	RunPod (Community)	Lambda Labs	Vast.ai (Median)
RTX 3090 24GB	$0.22/hr	—	$0.15/hr
RTX 4090 24GB	$0.34/hr	—	$0.43/hr
A100 SXM 80GB	$1.39/hr	$2.06/hr (8×)	$0.77/hr
H100 SXM 80GB	$2.69/hr	$3.44/hr (8×)	$1.91/hr
H200 141GB	$3.59/hr	—	$2.52/hr
B200 192GB	$5.98/hr	$5.74/hr (8×)	$2.67/hr

⚠️ Neither RunPod nor Lambda list the A10G specifically—both offer the A10 (24GB). Vast.ai prices are marketplace-driven and fluctuate minute-to-minute; computeprices Vast AI budget 30–50% above lowest advertised rate for predictable costs. GPUnex All pricing data sourced from provider pages and ComputePrices.com as of March 17–18, 2026.

5.2 New providers and Blackwell arrival

CoreWeave (IPO'd March 2025, ~$138/share) is the leading "neocloud" ABI Research with $5B+ projected annual revenue and major contracts with Microsoft Network World ($17.4–19.4B), Meta ($3B), MLQ and OpenAI. **Nebius** ("Europe's CoreWeave," spun from Yandex) raised $4.2B and signed a $27B Meta deal, Tech Funding News offering H200 at $3.50/hr and H100 at $2.95/hr. Network World **Crusoe** (sustainability-focused, ABI Research $10B+ valuation) runs on renewable energy. Runpod **DigitalOcean GPU Droplets** (formerly Paperspace) offers H100/H200 at $0.76–$7.99/GPU/hr with per-second billing. DigitalOcean Budget options include **SF Compute** (H100 at $1.45/hr), Saturn Cloud **Thunder Compute** (H100 at $1.38/hr), Thundercompute and **Vultr** (B200 at $2.89/hr reserved, MI300X at $1.85/hr). Saturn Cloud

NVIDIA Blackwell GPUs (B200, B300) are now available on RunPod, Lambda, Vast.ai, and others at $5–$6.40/hr, expected to push H100 pricing down another 10–20%. Introl AMD MI300X (192 GB) is gaining traction at $1.85/hr on Vultr as ROCm compatibility with PyTorch/JAX improves. Saturn Cloud

5.3 What it takes to serve Llama 4 Scout locally

Despite only 17B active parameters, **all 109B parameters must reside in memory** because any expert might be needed per token. EffiFlow Scout is bandwidth-bound, not compute-bound—VRAM Insiderllm capacity matters more than GPU compute speed.

- **Recommended datacenter setup**: 1× H100 80GB with Q4 quantization (~54.5 GB weights + KV cache)
- **Full precision**: 4× H100 (216 GB in BF16)
- **Consumer GPU (RTX 4090)**: Feasible with 1.78-bit Unsloth dynamic GGUF at ~20 tok/s (requires ~24 GB VRAM + 64 GB system RAM for CPU offloading)
- **Mac M3 Ultra (128 GB unified memory)**: ~47 tok/s at 4-bit quantization—one of the best consumer options due to massive unified memory and high bandwidth

Maverick requires approximately **3× the resources** of Scout (245 GB in Q4_K_M). For most local deployment scenarios, **Llama 3.3 70B or Qwen 3 32B deliver better results per dollar of VRAM** than Scout.

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6. Vector databases, RAG, and document infrastructure

6.1 Vector database landscape matures with clear tiers

The market has segmented into clear tiers. **Milvus** (~43,000 stars, Zilliz v2.6.11) Milvus leads at enterprise/billion-scale with a v3.0 roadmap for late 2026 featuring streaming ingestion and Vector Lake. Milvus **Qdrant** (~29,600 stars, v1.17.x) raised a **$50M Series B** (March 12, 2026) and replaced RocksDB entirely with its custom Gridstore engine. **Chroma** (26,000+ stars, Chroma v1.5.5) reached the 1.0 milestone with a new Rust execution engine and launched Chroma Cloud (serverless). **Weaviate** (~15,800 stars, v1.36.5) added Weaviate Agents and managed embedding services. **pgvector** (v0.8.2) achieved production-readiness for moderate scale (<5–10M vectors) with iterative index scans delivering up to **5.7× query performance improvement**.

A fundamental shift is underway: **vectors are becoming a data type, not a database type**. Oracle, Google, MongoDB, Elasticsearch, and every major database now supports vector operations, putting pressure on purpose-built vector databases. All major vector DBs are adding MCP support and agentic workflow integrations in response.

6.2 RAG evolves into agentic retrieval

RAG remains essential despite 1M+ token context windows. The consensus is clear: long-context windows are too expensive for production at scale, lack real-time data access, and don't provide the source attribution that enterprise compliance demands. Per Firecrawl (2026): "RAG is still very much relevant, as it is the best technique to enhance LLM's capabilities regardless of their size and context window."

The dominant patterns have shifted to **GraphRAG** (entity-relationship graphs enabling multi-hop reasoning, now mainstream since Microsoft's 2024 open-source release) and **Agentic RAG** (autonomous agents planning multi-step retrieval, choosing tools, and critiquing outputs). Self-RAG (models deciding when to retrieve), multi-agent RAG (parallel retrieval across different systems), and deep search (extended iterative refinement) are now standard approaches. RAGFlow describes the evolution as RAG transforming from "Retrieval-Augmented Generation" into a **"Context Engine"** with intelligent retrieval at its core.

6.3 Docling and LlamaIndex anchor document processing

Docling (IBM/LF AI & Data Foundation, **37,000+ stars**, MIT License) is now the leading open-source document parsing tool. Production-ready and deployed in banking via the Docling OpenShift Operator with Red Hat, it supports PDF, DOCX, PPTX, XLSX, HTML, images, audio, video, LaTeX, and financial reports. The **Granite-Docling-258M** vision-language model (Apache 2.0) performs single-pass extraction using DocTags markup. The ecosystem includes docling-serve (REST API), docling-java (enterprise), and docling-mcp (agentic workflows).

LlamaIndex (~47,764 stars) has pivoted from a RAG framework to an **agentic document processing platform**, now describing itself as "the leading document agent and OCR platform." Key products include LlamaParse v2 (cloud document parsing with up to 50% cost reduction), LlamaAgents (one-click deployment), LlamaExtract (structured data extraction), and event-driven Workflows for async agent orchestration. The common enterprise pattern uses LlamaIndex for document ingestion/parsing while using LangGraph for production orchestration.

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7. Chat UIs converge on agent platforms

Open WebUI dominates the self-hosted space with **127,665 stars**, **290 million+ downloads**, and 353K+ community users. It supports Ollama, OpenAI-compatible APIs, and Anthropic backends with advanced RAG across **9 vector database options** (ChromaDB, pgvector, Qdrant, Milvus, Elasticsearch, OpenSearch, Pinecone, S3Vector, Oracle 23ai). Major features include a built-in code interpreter, voice/video calls, Python function calling, persistent artifact storage, enterprise features (SSO, RBAC, SCIM, audit logs), and OpenTelemetry observability. A desktop app is in alpha.

LibreChat (~34,300 stars, v0.8.3-rc1) was **acquired by ClickHouse** in 2025 and targets enterprise teams wanting a ChatGPT alternative. It offers full agent support with MCP integration, Code Interpreter, Artifacts (generative UI with Monaco editor), and Responses API support. Enterprise users include Daimler Truck. The 2026 roadmap includes an admin panel (GUI-based configuration replacing YAML editing) and interactive workflows with human-in-the-loop.

LobeChat (~50,000+ stars, recently rebranded to LobeHub) takes an agent-first approach with multi-agent collaboration, MCP Marketplace, knowledge base/RAG, and a notably polished modern UI. It offers desktop apps and one-click Vercel deployment. **NextChat** remains popular for lightweight, cross-platform use (~5 MB client).

The defining trend since mid-2025 is that chat UIs are becoming **agent platforms**—MCP integration, tool calling, code execution, and multi-agent orchestration are now table-stakes features rather than differentiators.

Conclusion

Five structural shifts define the March 2026 open AI infrastructure landscape. First, **MoE has become the dominant architecture** for frontier open-weight models—Llama 4, Mistral Large 3, Qwen 3, DeepSeek-R1, and GPT-OSS all use mixture-of-experts, delivering frontier performance with dramatically fewer active parameters per token. Second, **MCP is the emerging**

interoperability standard, now a Linux Foundation project with 97 million SDK downloads and 10,000+ servers, though its security model remains immature and no inference engine natively implements it. Third, **agent frameworks have converged** around common patterns—graph-based workflows, MCP tool integration, human-in-the-loop, and multi-agent orchestration—with Microsoft, Google, OpenAI, Hugging Face, and LangChain all offering mature solutions. Fourth, **GPU cloud pricing has collapsed** 64–75% from peak, with H100s now available under $2/hr on marketplace providers, making local and cloud model serving dramatically more accessible. Fifth, **RAG has evolved into agentic retrieval** rather than being replaced by long-context models—GraphRAG and multi-step agentic patterns are now the production standard.

For the ContextKeeper whitepaper, the most significant gaps in the current document are the absence of MCP (now foundational open infrastructure), the outdated Llama 3.1 flagship reference (Scout/Maverick represent a fundamentally different architecture), and the separate treatment of AutoGen and Semantic Kernel (now unified in Microsoft Agent Framework). The addition of GPT-OSS, Qwen 3, and SGLang would also substantially strengthen the technical reference.